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Mud Invasion Zone Detection Using Retrieval-Augmented Generation (RAG): A Generative AI Approach

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Abstract

Mud invasion, the process where drilling mud infiltrates surrounding rock formations, poses a significant challenge in well log interpretation within the petroleum industry. This phenomenon distorts measurements of formation properties such as porosity, permeability, and fluid content, leading to inaccuracies in formation evaluation and hydrocarbon volume estimation. Traditionally, detecting mud invasion zones requires manual analysis of well log data, particularly resistivity, bulk density, neutron, and sonic logs which are time-consuming and subject to interpretation biases. This study presents an automated approach to identifying mud invasion zones in well logs using Retrieval-Augmented Generation (RAG) with advanced indexing techniques, specifically Vector Store Index and Tree Index RAG.

Our approach integrates a comprehensive RAG pipeline that includes naive RAG, Vector Store Index, and Tree Index RAG, leveraging domain-specific manuals and well data stored in LAS format, transformed into text files through a custom processing pipeline. These technical resources, including annotated "bad data zones" indicating mud invasion, enable the model to contextualize well log anomalies with greater precision. Additionally, the RAG system not only detects mud invasion zones but also retrieves relevant guidance from manuals on handling these cases, enhancing the model's practical application in real-world scenarios.

To evaluate performance, we use Cosine Similarity based on TF-IDF (Term Frequency-Inverse Document Frequency), ROUGE Score, Human Feedback (HF), and processing time metrics. TF-IDF allows Cosine Similarity to measure the relevance of retrieved information against manuals, while ROUGE Score assesses the overlap between generated outputs and pre-annotated bad data zones, providing an accuracy measure. Results show that the Vector Store Index RAG configuration outperforms others in both processing time and accuracy metrics. It achieves higher relevance and accuracy and offers significantly reduced processing times compared to naive and Tree Index RAG configurations. This efficiency makes Vector Store Index particularly suitable for practical deployment in well log analysis.

Automating mud invasion detection can streamline workflows, reduce human error, and ensure consistency in well log analysis. This study underscores the growing role of Generative AI in the energy sector, where vast amounts of unstructured data must be processed efficiently. Our research demonstrates how combining NLP and domain knowledge can improve well log interpretation, leading to more accurate and scalable reservoir assessments. This work represents a step toward integrating AI-driven solutions in subsurface characterization, advancing automated interpretation in the energy sector.

Keywords: Retrieval-Augmented Generation (RAG), Mud Invasion Detection, Well Log Interpretation, Vector Store Indexing, Tree-Based Retrieval, Generative AI

Introduction

Accurate detection of the mud invasion zone is a critical task in formation evaluation. The mud filtrate invasion that occurs during drilling can significantly alter the petrophysical properties measured by logging tools, leading to misinterpretation of true rock and fluid properties. Identifying the depth and extent of the invaded zone enables more reliable estimates of original formation resistivity, porosity, fluid saturation, and hydrocarbon potential (Eltsov et al. 2011). This is particularly important in complex reservoirs, where insignificant variations in invasion profiles can affect decisions on perforation, completion design, and reservoir modeling (LIU et al. 2012). Traditional approaches for detecting mud invasion, such as resistivity log interpretation, require domain expertise and often rely on simplified analytical models or empirical assumptions, which may not generalize well across heterogeneous formations (Ribeiro and Carrasquilla 2013).

With the growing availability of unstructured data in the energy sector, there is an increasing interest in using advanced AI techniques to derive insights that were previously hidden or underutilized. One promising approach is Retrieval-Augmented Generation (RAG), a hybrid framework that combines the strengths of large language models (LLMs) with external knowledge retrieval mechanisms. Originally developed to enhance performance on open-domain question answering tasks, RAG enables the integration of both structured and unstructured domain-specific information to generate contextually accurate and knowledge-rich responses (Gao et al. 2023).

Recent applications for RAG in the energy sector have started to emerge. As an example, (Zhang, Li, and Su 2025) demonstrated that combining LLM with RAG enables the automatic generation of well logging interpretation reports with expert-level clarity and efficiency, reducing preparation time from hours to minutes. As another example, RAG showed an improvement in answering drilling-specific queries compared to baseline models, highlighting its value in extracting operational insights from unstructured text data in the petroleum domain (Liang, Zhou, and Gurbani 2024). By incorporating context-aware retrieval from geological databases and technical documentation, RAG offers a path forward for intelligent interpretation workflows that go beyond pattern recognition capable of reasoning, explanation, and integration across diverse data sources.

In this study, we present an automated approach for identifying mud invasion zones in well logs using Retrieval-Augmented Generation (RAG) with advanced indexing methods, namely Vector Store Index and Tree Index RAG. Focusing on two key scenarios of mud invasion zone such as resistivity log comparison and spontaneous potential (SP) analysis, our system retrieves relevant prior cases and domain knowledge to generate context-specific interpretations via a generative language model. This work advances human-in-the-loop AI for petrophysical analysis, combining data-driven insights with historical expertise.

Retrieval-Augmented Generation (RAG) and Indexing Methods

RAG systems operate by combining two core components: a retriever and a generator. The retriever identifies relevant document chunks from a knowledge base in response to a user query, and the generator subsequently uses this information to produce a coherent, contextually grounded response (Şakar and Emekci 2025). This hybrid framework addresses a key limitation of standalone generative models, namely, the risk of hallucination by anchoring outputs in retrieved content.

In petroleum geoscience and drilling analytics, where domain-specific information such as annotated well logs, drilling manuals, and operational reports are essential, RAG architectures provide a mechanism for incorporating this external knowledge into automated systems. The efficacy of a RAG system, however, is highly dependent on the design of its retrieval component. In this study, we investigate three retrieval configurations: Naive RAG, Vector Store Index RAG, and Tree Index RAG.

Naive RAG Baseline

As a comparative baseline, we also consider Naive RAG, where retrieval is conducted via standard dense search without specialized indexing or embedding optimization. While simpler to implement, this method often lacks the precision necessary for domain-specific applications, particularly when semantic alignment between query and document is subtle. Naive RAG approaches have been shown to underperform in accuracy and efficiency relative to their indexed counterparts, particularly on domain-specific benchmarks (Pratama et al. 2025)

Vector Store Index RAG

Vector Store Indexing leverages dense embeddings of document content to enable semantic search within a vector database. Each document chunk is transformed into a high-dimensional vector representation using pre-trained embedding models such as OpenAI's text-embedding-ada-002 or BAAI's bge-small-en, which have been optimized for retrieval performance in domain-specific applications (Tahseen et al. 2025). These vectors are stored using databases like FAISS (Johnson et al., 2017) or DeepLake, allowing similarity-based retrieval using cosine distance.

This approach is especially suited for structured domains where precision and retrieval speed are critical. Prior work has demonstrated that embedding-based search significantly outperforms traditional keyword-based methods in terms of both recall and response relevance, particularly in technical disciplines such as medicine(Tahseen et al. 2025) and engineering. In the context of well log analysis, embedding-based retrieval facilitates the alignment of queries such as "indicators of mud invasion" with highly specific and context-rich document segments. The architecture overview was illustrated in Figure 1.

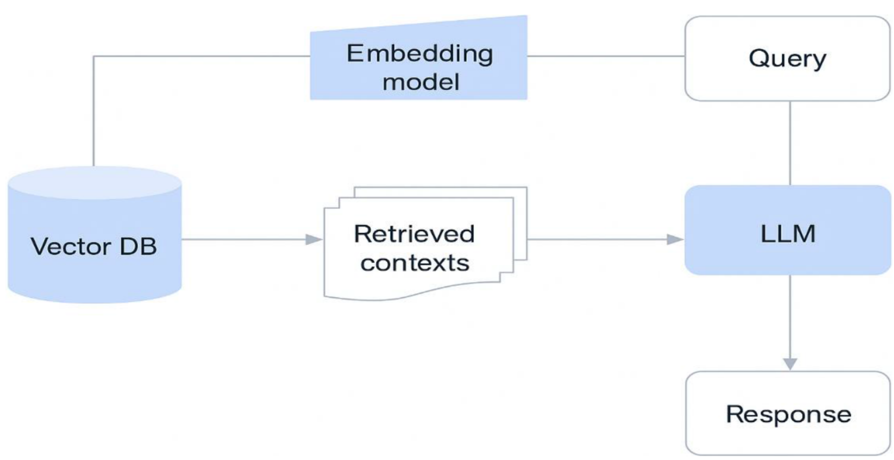

Figure 1—Workflow of a Vector Store-Based Retrieval-Augmented Generation (RAG) System

Tree Index RAG

Tree-based indexing, implemented in systems like LlamaIndex, introduces a hierarchical structure over the document corpus. In this framework, individual text chunks are recursively summarized to form internal nodes, resulting in a tree architecture where each node encapsulates progressively broader semantic content (Yang and Huang 2025). At query time, the retrieval process begins at the root and narrows down through child nodes using relevance scoring, allowing for more context-aware traversal and retrieval shown as in Figure 2:

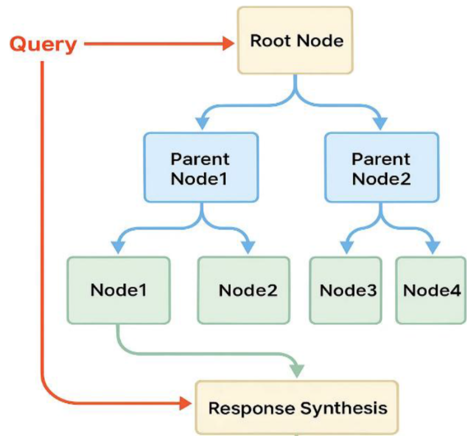

Figure 2—Tree Index-based retrieval and response synthesis flow

This hierarchical approach is particularly beneficial in scenarios where relevant information is distributed across multiple sections or when queries require reasoning over summarized knowledge. Tree indexing has been explored in scientific literature summarization (Rothman 2024) and legal document reasoning (Asai et al. 2023), where layered knowledge is critical for effective inference. However, this method typically incurs greater computational overhead due to recursive lookups and summarization steps.

Methodology

To develop a Retrieval-Augmented Generation (RAG) framework capable of detecting mud invasion zones within well logs, we designed a multi-stage methodology that integrates data transformation, text indexing, and generative modeling. The process encompasses both structured well log data in LAS format and unstructured domain knowledge extracted from technical manuals. Figure 3 outlines the step-by-step approach we used to build and evaluate three RAG configurations: Naive, Vector Store, and Tree Index.

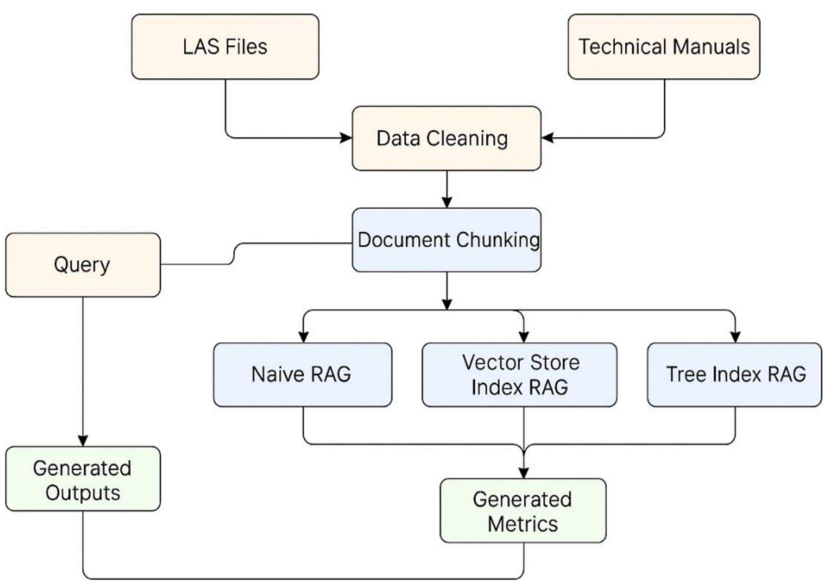

Figure 3—General overview of methodology for RAG systems

Data Preparation and Preprocessing

Our dataset comprised two primary sources: well log data and petroleum engineering manuals. Structured data consisted of digital well logs stored in LAS (Log ASCII Standard) format, each containing 1,617 depth-indexed data points across multiple channels such as resistivity, porosity, caliper, and density. In parallel, unstructured domain knowledge was gathered from a variety of technical manuals, including vendor guidelines, academic literature, and internal training documents. Additionally, a custom-skewed manual was curated by summarizing domain-specific diagnostic rules and covering a wide range of data point scenarios relevant to mud invasion, typically ranging from 1,450 to 1,615 meters. This diverse data foundation enabled the construction of a retrieval system capable of context-aware generation and grounded decision support. These files were parsed using a custom Python script that extracted numerical values and converted them into plain text format, preserving depth intervals for later reference. In parallel, technical manuals were also cleaned and formatted into structured, readable text.

We then performed standard data preprocessing: removing redundant headers, normalizing units, eliminating irrelevant characters, and ensuring consistent casing. Particular attention was paid to aligning known mud invasion zones previously annotated by experts with corresponding log intervals to facilitate supervised evaluation later in the workflow.

Text Chunking and Embedding

To prepare both logs and manuals for retrieval, all documents were segmented into overlapping text chunks of approximately 200-300 words using a sliding window approach. This chunking ensured that context was preserved across boundary regions and that important log behaviors (e.g., sudden drops in resistivity) were not lost due to segmentation.

Each chunk was embedded into a high-dimensional semantic space using the BAAI/bge-small-en model, selected for its strong performance in retrieval-focused tasks. The resulting embeddings were stored in a DeepLake vector store, enabling efficient similarity searches based on cosine distance.

RAG Configurations and Retrieval Strategies

We implemented three retrieval strategies to compare their effectiveness in this domain:

- In the Naive RAG setup, document chunks were retrieved using basic keyword similarity without vector embeddings or structural indexing. This served as our baseline for performance comparison.
- The Vector Store Index RAG approach relied on the DeepLake vector database to perform semantic retrieval using cosine similarity. Top-k relevant chunks were retrieved and passed to a generative model (OpenAI GPT-4.0), which used this content to generate a domain-grounded response.
- In the Tree Index RAG variant, documents were organized hierarchically using a recursive summarization process implemented via LlamaIndex. The retrieval mechanism traversed this tree structure from root summaries to detailed child nodes, collecting and passing only the most contextually aligned content to the generator.

Each configuration was integrated into a LlamaIndex pipeline that allowed consistent formatting of queries, retrieval parameters, and generative prompts.

Query Design

To evaluate the practical applicability of the proposed RAG configurations, we formulated a set of domain-specific queries that reflect realistic analytical challenges encountered during drilling and well log interpretation. These queries were designed to assess each system's ability to detect indicators of mud invasion, interpret resistivity anomalies, and extract relevant diagnostic procedures from technical manuals.

Examples include: "Is there any mud invasion zone in the well?", "What is the depth interval where mud invasion occurred?", and "How did you identify mud invasion zones?" Each query was written in plain, operational language to simulate how a practicing petrophysicist or drilling engineer might interact with an AI assistant. The same query set was applied uniformly across all three configurations—Naive RAG, Tree Index RAG, and Vector Store Index RAG—to enable consistent evaluation under identical input conditions.

Evaluation Metrics

To comprehensively assess the performance of each RAG configuration (Naive, Vector Store Index, and Tree Index), we employed a multi-metric evaluation framework that includes both quantitative and qualitative metrics. These metrics were selected to capture semantic accuracy, lexical overlap, human interpretability, and computational efficiency. The following criteria were used:

Cosine Similarity (TF-IDF Weighted):

To assess the relevance of retrieved documents to domain-specific queries (Juvekar and Purwar 2024), we computed cosine similarity between the TF-IDF vector of the user query and each retrieved chunk using Eq. 1:

$$\mathit{Cosine\ Similarity}\,(A, B) = \frac{A \cdot B}{\| A \| \, \| B \|} \tag{1}$$

where A is the TF-IDF vector representation of the query, and B is the TF-IDF vector of a retrieved passage. This metric quantifies the angular distance between the vectors, with values closer to 1 indicating high semantic similarity. The final score is computed as the average similarity across all top-k retrieved documents.

ROUGE Score (Recall-Oriented Understudy for Gisting Evaluation)

ROUGE scores were used to compare generated responses to manually annotated "bad data zones" extracted from the well logs. We specifically used ROUGE-1 (unigram overlap) and ROUGE-L (longest common subsequence) as defined in Eq. 2 and Eq. 3 respectively:

ROUGE-1 Recall:

$$ROUGE-1 = \frac{\mathit{Overlapping\ Unigrams}}{\mathit{Total\ Reference\ Unigrams}} \tag{2}$$

ROUGE-L (based on LCS):

$$ROUGE-L = \frac{\mathit{Length\ of\ LCS}}{\mathit{Reference\ Length}} \tag{3}$$

These metrics quantify the lexical overlap between system outputs and annotated ground truth answers (Lin 2004).

Human Feedback (Expert Rating)

Two petrophysics domain experts (authors) independently evaluated the responses generated by each RAG configuration. They scored each output on three criteria:

- Relevance (Does the response address the query appropriately?)
- Accuracy (Is the explanation consistent with domain knowledge?)
- Actionability (Can the response be used to guide decisions?)

Each criterion was rated on a 5-point Likert scale, where 1 = "Poor" and 5 = "Excellent." The final As suggested in literature (Bai et al. 2024), Human Feedback score was computed as the average across raters and criteria per RAG configuration.

Processing Time (Latency per Query)
We measured the average time (in seconds) required by each RAG variant to generate a response, from query input through document retrieval to final output generation. This metric was captured over 20 trials per method and is reported in Eq. 4:

$$Avg.Processing\ Time = \frac{1}{n}\sum_{i=1}^{n} t_i \tag{4}$$

Where t_i is the time recorded for the i_{th} query. Lower processing time indicates better scalability and suitability for real-time decision-making scenarios.

Results

Figure 4 illustrates one of the well data under examination. According to petrophysical analysis, the lower part of the interval suggests the presence of hydrocarbon-bearing zone. Potential mud invasion zones were also found by petrophysical experts in that depth interval (at around 1580 m), in addition to mud invasion zones found in the middle of the well data interval.

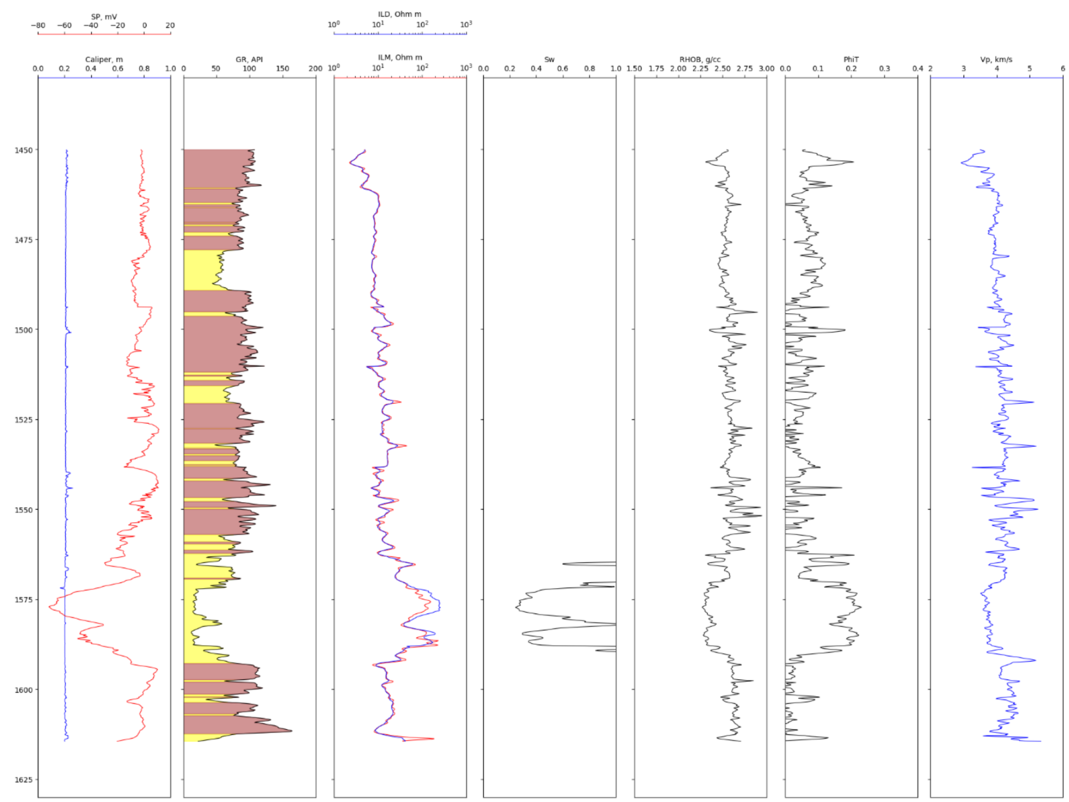

Figure 4—Example of well data. From left to right: caliper, spontaneous potential (SP), deep and medium resistivity, water saturation (Sw), bulk density (RHOB), porosity (PhiT), P-wave velocity (Vp) logs.

To assess the performance of the three RAG configurations, namely Naive RAG, Vector Store Index RAG, and Tree Index RAG, we conducted a comparative evaluation using domain-specific queries and multiple metrics, including cosine similarity, ROUGE scores, human feedback, and processing time.

Cosine Similarity (TF-IDF)

Among the three methods, the Vector Store Index RAG configuration achieved the highest cosine similarity scores, indicating that the retrieved chunks were more semantically aligned with the queries. On average, it scored 0.84, compared to 0.72 for Tree Index RAG and 0.63 for Naive RAG. This confirms the superior semantic retrieval capability of the embedding-based vector search approach.

ROUGE Scores

The generated outputs were compared to expert-annotated mud invasion descriptions using ROUGE-1 and ROUGE-L metrics. Vector Store Index RAG again outperformed, achieving an average ROUGE-1 recall of 0.61 and ROUGE-L of 0.58. In contrast, Tree Index RAG yielded ROUGE-1 and ROUGE-L scores of 0.53 and 0.49, respectively, while Naive RAG trailed with 0.42 and 0.39. These results suggest that the outputs generated by Vector Store Index RAG more closely matched the lexical patterns and content of expert labels. A summary of the quantitative evaluation metrics, including cosine similarity and ROUGE scores across all RAG configurations, is presented in Table 1:

Table 1—Quantitative Metrics Comparison

Metric	Naive RAG	Tree Index RAG	Vector Store RAG
Cosine Similarity	0.63	0.72	0.84
ROUGE-1	0.42	0.53	0.61
ROUGE-L	0.39	0.49	0.58

Human Feedback (Expert Ratings)

Two domain experts independently evaluated the responses from each RAG configuration using three criteria: relevance, accuracy, and actionability. As summarized in Table 2, the Vector Store Index RAG achieved the highest average rating of 4.5 out of 5, followed by Tree Index RAG with 4.0, and Naive RAG with 3.0. Expert reviewers described the outputs generated by the Vector Store configuration as "directly usable" and "consistent with known invasion patterns," particularly in identifying depth-specific anomalies across resistivity and porosity logs. These findings highlight the configuration's practical reliability in supporting well log interpretation.

Table 2—Expert Evaluation of RAG Configurations on Mud Invasion Detection Queries

Prompt	Naive RAG Response	Tree Index RAG Response	Vector Store Index RAG Response
1. Is there any mud invasion zone in the well?	"There are no strong signs of mud invasion based on the available logs. The SP and density logs appear normal."	"There may be signs of mud invasion, especially where resistivity readings deviate, but further analysis is needed."	"Yes, the analysis of resistivity and caliper logs reveals clear signs of mud invasion, particularly where the resistivity drops and the borehole diameter increases."
2. What is the depth interval where mud invasion occurred?	"Mud invasion could be occurring somewhere above 1525 m, but the exact depth is difficult to determine."	"A potential mud invasion zone is observed between 1500 m and 1550 m based on resistivity fluctuations."	"The mud invasion is most prominent between 1525 m and 1590 m, where resistivity logs show distinct separation and caliper log indicates borehole enlargement."
3. How did you identify mud invasion zones?	"The invasion might be inferred from sudden changes in the SP and density logs, though other factors could also cause this."	"By analyzing changes in caliper and resistivity logs, which could suggest fluid invasion into the formation."	"By comparing the resistivity readings from SFL, ILM, and ILD logs. The mud filtrate has a different electrical resistivity than the original formation fluids (oil, water, gas). This difference results in characteristic log responses that indicate the presence of mud invasion."

Prompt	Naive RAG Response	Tree Index RAG Response	Vector Store Index RAG Response
Expert Review Summary	Lacks specificity and often uses incorrect log indicators; reasoning is too general.	Reasonable log selection with moderate clarity, but explanations could be more precise.	Consistently uses correct logs, delivers clear, technically sound, and actionable interpretations.
Average Score	**3.0/5.0**	**4.0 / 5**	**4.5 / 5**

Processing Time

Processing time was a critical factor in evaluating the practicality of each RAG configuration. While the Vector Store Index RAG exhibited the fastest performance with an average of 3.2 seconds per query, the Tree Index RAG, despite achieving higher accuracy than the Naive RAG, incurred the highest latency at 5.7 seconds. This added delay is attributed to its recursive traversal and summarization mechanism. The Naive RAG, by contrast, required 4.8 seconds on average but produced lower-quality outputs. These results suggest that although Tree Index RAG offers improved interpretability over Naive RAG, its response time may be a limiting factor in real-time applications. A visual comparison of processing times for all three configurations is presented in Figure 5. Although seemingly small, a 1–2 second reduction per query translates to substantial time savings when scaled to thousands of real-time queries during drilling operations, thereby enhancing the responsiveness and practical viability of the system.

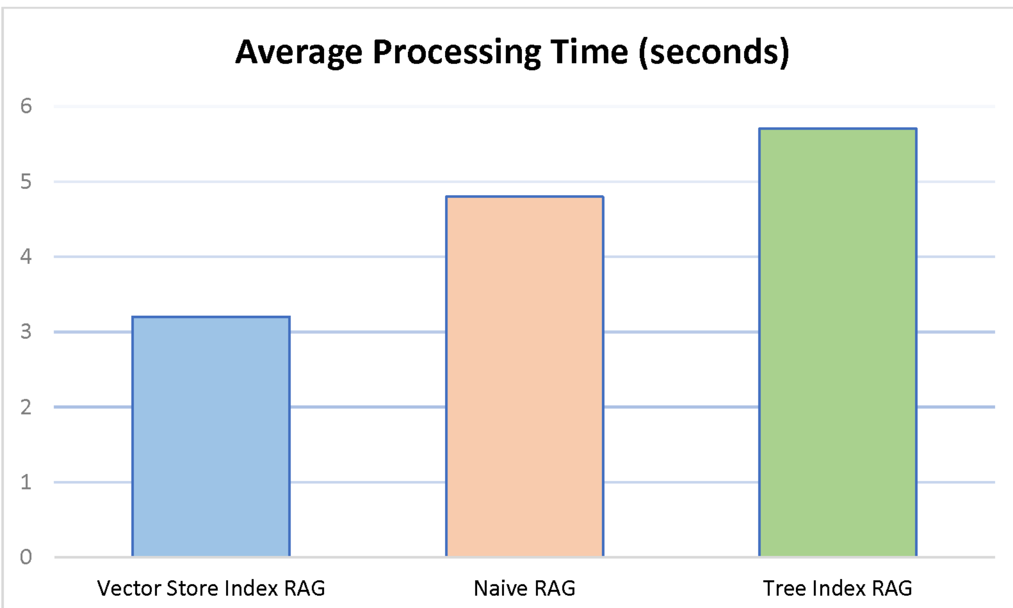

Figure 5—Average Processing Time per Query for RAG Configurations

Conclusions

This study presents a novel application of Retrieval-Augmented Generation (RAG) for automating the detection of mud invasion zones in well logs, a task that traditionally demands expert interpretation and is prone to subjectivity. By integrating domain-specific knowledge from technical manuals and well log data, we evaluated three RAG configurations, such as Naive RAG, Tree Index RAG, and Vector Store Index RAG, using both quantitative metrics and expert feedback.

Our results demonstrate that the Vector Store Index RAG significantly outperforms the other configurations in terms of retrieval accuracy, response relevance, and processing efficiency. It consistently generated more accurate and actionable responses while maintaining low latency, making it a practical solution for real-time decision support during drilling operations.

The success of this approach underscores the transformative potential of Generative AI in the energy sector, specifically for tasks involving unstructured data interpretation. By combining natural language processing with domain-specific embeddings and structured retrieval pipelines, RAG systems can replicate expert-level reasoning and provide scalable solutions for subsurface characterization. Ultimately, this research marks a step toward building intelligent, human-in-the-loop systems that can augment petrophysical workflows and accelerate data-driven decision-making in reservoir evaluation.

Nomenclature

RAG	Retrieval-Augmented Generation
LAS	Log ASCII Standard (well-log file format)
TF-IDF	Term Frequency–Inverse Document Frequency
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
HF	Human Feedback (expert rating)
FAISS	Facebook AI Similarity Search (vector database library)
GPT-4	Generative Pre-trained Transformer, version 4
SP	Spontaneous Potential log
RHOB	Bulk Density log
SFL	Shallow Focused Log (shallow resistivity)
ILM / ILD	Medium and Deep Laterolog resistivity logs, respectively

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